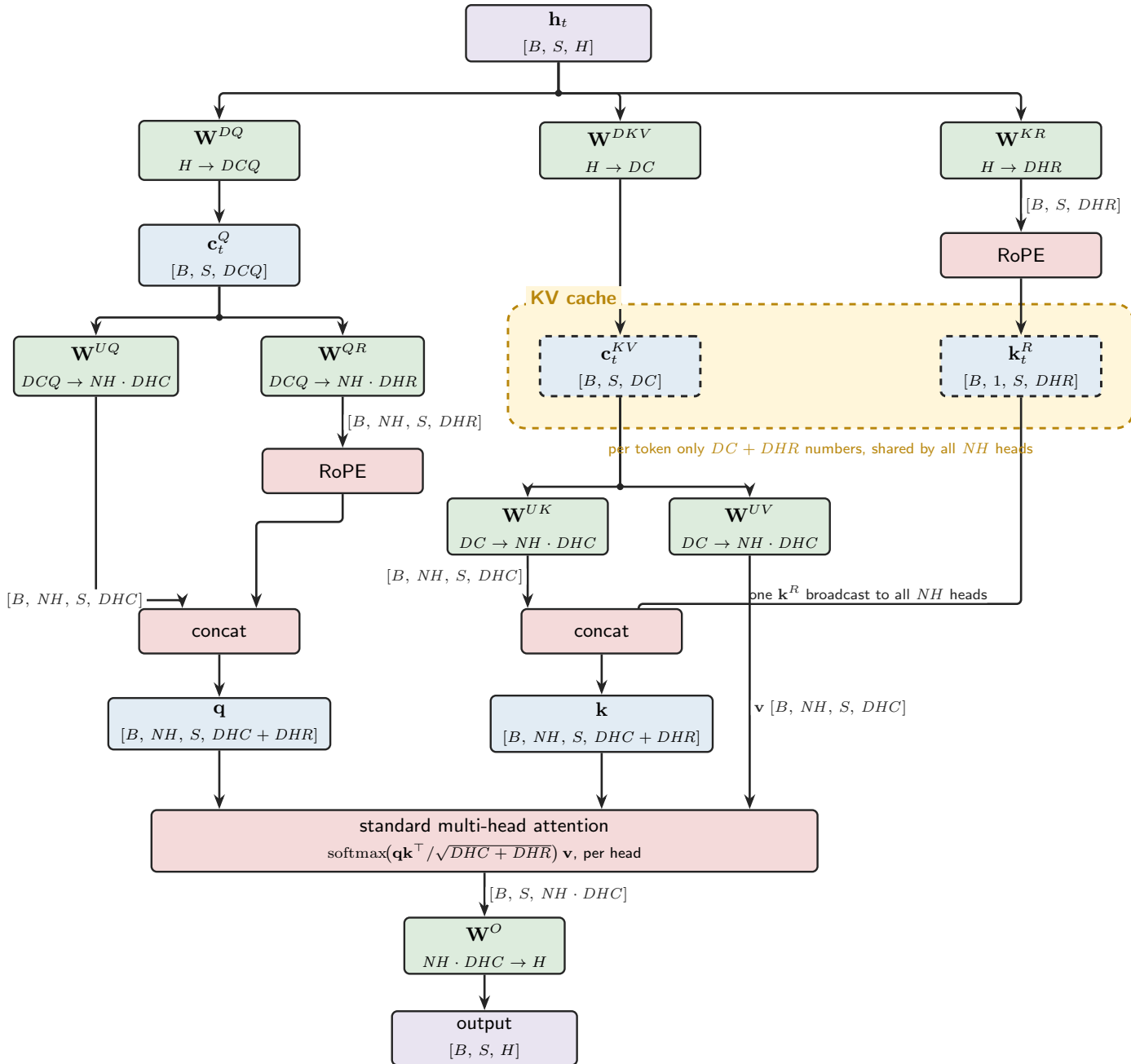


Multi-head Latent Attention (MLA): low-rank Q / KV, decoupled RoPE, and what the KV cache holds



H hidden, NH heads, DHC content dim per head, DHR rotary dim per head, DC KV latent dim, DCQ query latent dim.

DeepSeek-V3: $H=7168$, $NH=128$, $DHC=128$, $DHR=64$, $DC=512$, $DCQ=1536$. Green = weight, blue = activation, pink = operator.